ALBERT — The Ultimate AP Chemistry Cheat Sheet (v2)

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1-2: Atomic & Molecular Structure (14–18%)

Electron Config & Periodic Trends

Aufbau: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p... **Hund:** single-fill each orbital before pairing. **Pauli:** max 2 e⁻/orbital (opposite spins). **Periodic trends** (L→R, top→bot): radius ↓R, ↑down. IE ↑R, ↑up. EA ↑R (dips at N, O). EN ↑R, ↑up.

Ions: +e⁻ anion (> parent), -e⁻ cation (< parent). **Shielding:** inner e⁻ repel outer →↓ Z_{eff} → binding E ↓, radius ↑.

Isoelectronic: same e⁻ count; smaller radius w/ ↑ nuclear charge.

PES: peak position = binding E; peak height ∝ # e⁻.

Coulomb: $F \propto q_1q_2/r^2$, $E \propto q_1q_2/r$. (q = charge, r = distanc.)

Lewis Structures, Formal Charge, Polarity

Steps: sum valence e⁻; skeleton w/ least-EN atom central (never H); fill octets; multiple bonds if short.

Formal Charge (FC) $= V - N - B/2$ (V=val, N=lone, B=bond e⁻); best structure has IFCI minimized; −FC on more-EN atom.

Octet exceptions: H = 2; Be = 4; B/Al = 6; expanded octets for period 3+ (P, S, Cl); odd-e⁻ (NO, NO_2).

Resonance: same atoms, different e⁻ position; real = weighted avg. All resonant bonds are identical (intermediate length/energy)

VSEPR Geometry & Hybridization

e⁻ grps	Bonds	LP	Shape	Angle	Hybrid
2	2	0	linear	180°	sp
3	3	0	trig planar	120°	sp^2
3	2	1	bent	$< 120^\circ$	sp^2
4	4	0	tetrahedral	109.5°	sp^3
4	3	1	trig pyr	$< 109.5^\circ$	sp^3
4	2	2	bent	$\ll 109.5^\circ$	sp^3

✓ Polar molecule = polar bonds + asymmetric shape. EN diff: > 1.7 ionic; 0.4–1.7 polar cov; < 0.4 nonpolar.

3: Intermolecular Forces & Properties (18–22%)

✓ IMF tree: H–F/O/N? → H-bond. Charged? → ion-dipole. Polar? → dipole-dipole. Else → LDF. List strongest first.

IMF Types & Physical Properties

Ranking: ion-dipole > H-bond > dipole-dipole > LDF. **H-bonds:** H–F/O/N (DNA, ice, water — abnormally high bp). **LDF:** in ALL molecules; ↑ w/ surface area, e^- count, polarizability (enhanced by π bonding). **Trends:** ↑ IMF → ↑ bp/mp/viscosity/surface tension, ↓ P_{vap}. Ice < water density (H-bonds). **"Like dissolves like."** Polar + ionic in H_2O; nonpolar in hexane/oil.

Gases & KMT

KMT: point particles, random motion, elastic collisions, no IMFs, $KE \propto T$. Same T = same avg KE for all gases. Heavier = slower. **Max-Boltz:** ↑ T or ↓ mass = wider, right-shifted curve.

Ideal gas: $PV = nRT$ (R = 0.0821 L·atm/mol·K). **Combined:** $P_1V_1/T_1 = P_2V_2/T_2$. **STP:** 0°C, 1 atm → 22.4 L/mol.

Dalton: $P_{tot} = \Sigma P_i$; $P_i = \chi_i \cdot P_{tot}$ ($\chi_i = n_i/n_{tot}$).

KMT eq: $KE = \frac{1}{2}mv^2$. B/c avg KE depends only on T, lighter gases (smaller m) → higher avg velocities (effuse/diffuse faster).

Deviations: real $P <$ ideal (IMFs, worst low T); real $V >$ ideal (particle vol, worst high P). Ideal at high T + low P.

Solutions & Separations

Concentration: M = mol/L. **Dilution:** $M_1V_1 = M_2V_2$.

Solubility of solids ↑ w/ T; gases ↓ w/ T, ↑ w/ P. **Chromatog:** separates by IMF differences; furthest spot = highest mobile-phase affinity. **Distillation:** weakest IMFs (bp) boils first.

Photons, Spectroscopy & Beer-Lambert

Photon E = molecule's ΔE: $E = h\nu = hc/\lambda$. Micro = rot; IR = vib; UV/Vis = elec transitions. **Beer:** $A = \epsilon bc$ ($A \propto c$ if b, λ fixed). Set λ at max A for highest sensitivity.

Chemical Reactions (Unit 4, 7–9%)

Spectator ions cancel in net ionic. Cross same ions both sides.

Mole Conversions, Empirical & Molarity

Mole: $n = g/MW$; $g = n \cdot MW$; particles $= n \cdot 6.022 \times 10^{23}$. **% comp:** (mass element / total MW) $\times 100$

Empirical: divide mol of each element by smallest mol → simplest whole-num ratio (multiply to clear frac, e.g., x2 for 0.5).

Molecular formula = empirical $\times (MW_{actual}/MW_{empir})$.

Molarity: $M = n/V_L$. **Dilution:** $M_1V_1 = M_2V_2$.

Stoichiometry & Limiting Reagent

Workflow: balance → $g \to n \to$ mole ratio $\to n \to g$.

Limiting: compute moles product from EACH reactant; smallest wins. Excess = total − used. **% yield** = (actual/theor) $\times 100\%$.

Reaction Types

Synthesis: $A + B \to AB$. **Decomp:** $AB \to A + B$. **Single replace:** $A + BC \to AC + B$ (activity series). **Double replace:** $AB + CD \to AD + CB$ (driving force: precipitate, gas, weak electrolyte). **Combustion:** $C_xH_y + O_2 \to CO_2 + H_2O$; incomplete → CO.

Redox & Net Ionic

OS: element =0; ion = charge; O = -2 (peroxide −1); H =+1 (hydride -1); F =-1. Sum of OS in cmpd =0; polyat. ion = ovr chrg

Redox: OS ↑ = ox (loses e⁻); OS ↓ = red (gains e⁻).

Half-rxn: atoms (ex O/H) → H_2O for O → H^+ for H → charge w/ e^- → multiply to cancel e^- → combine. (Basic: neutralize H^+)

Net ionic: split aq ions (keep s/l) → cancel spectators.

Soluble: Group 1, NH_4^+, NO_3^-. **Insol:** OH^-, CO_3^{2-}, PO_4^{3-}, S^{2-} (ex Group 1); SO_4^{2-} w/ Ba/Pb/Sr/Ca.

✓ FRQ: balance → ionic → cancel → net w/ states. Redox: half-rxns → atoms/charge/e⁻ → combine.

Kinetics (Unit 5, 7–9%)

✓ Rate law from experiment, not balanced equation coefficients. Exception: For a single elementary step, coeff's = exponents.

Rate Laws & Order

rate $= k[A]^x[B]^y$ (x, y from expt). Overall order $= x + y$. k units: $M^{1-\text{order}}s^{-1}$. **Doubling [A]:** rate same → 0th; ×2 → 1st; ×4 → 2nd. **Relative Rates:** for $aA \to bB$, rate $= -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = \frac{1}{b}\frac{\Delta[B]}{\Delta t}$. Only 1st order $t_{1/2}$ is constant (e.g., radioactive decay). **Beer's Law:** Use A for [].

Order	Integ. Law	Linear Plot	$t_{1/2}$	k units
0th	$[A] = [A]_0 - kt$	$[A]$ vs t	$[A]_0/2k$	M/s
1st	\ln[A] = \ln[A]$_0$ - kt	$\ln[A]$ vs t	$0.693/k$	s^{-1}
2nd	1/[A] = 1/[A]$_0$ + kt	$1/[A]$ vs t	$1/(k[A]_0)$	$M^{-1}s^{-1}$

Activation Energy & Arrhenius

Arrhenius & E_a: Higher T or lower E_a → higher k (faster rate). Math calculations using Arrhenius are NOT tested.

Rxn coord diagram: E_a = peak − reactants; $\Delta H = E_{a,\text{fwd}} - E_{a,\text{rev}}$. Catalyst lowers both E_a, no ΔHchg.

Mechanisms & Collision Theory

RDS: slowest step sets rate; exponents from RDS molecularity. **Pre-equilibrium approx.:** If RDS has an intermediate, set fast step $rate_{fwd} = rate_{rev}$ to substitute out $[Int]$ in the rate law. **Catalyst:** in mechanism, NOT net rxn. **Intermediate:** made then used; never in overall. Steps sum to overall. **Collision theory:** need $E \geq E_a$ + correct orientat.. ↑ T ↑ *fraction* w/ sufficient E. **Maxwell-Boltzmann:** high T curve flattens & shifts right. Termolecular steps (3+ colliding) are rare.

✓ FRQ: Justify order from data (doubling/integrated); include k units. Justify: steps sum to overall & rate law matches RDS.

Thermodynamics (Unit 6, 7–9%)

Heat, Calorimetry & Sign Conventions

$q = mc\Delta T$ (c = specific heat, J/g·°C; $c_{H_2O} = 4.18$). $\Delta T = T_f - T_i$. Heat cap $C = mc \to q = C\Delta T$.

Signs: $q > 0$ system absorbs (endo); $q < 0$ releases (exo). Surroundings opposite. **Coffee cup** (const P): $\Delta T_{sol} \to q_{sol} = mc\Delta T \to q_{rxn} = -q_{sol} \to \Delta H_{rxn} = q_{rxn}/n$ (J/mol).

Mixing: $q_{lost} + q_{gained} = 0$. Isolated syst. → no heat lost to env.

Thermal eq: Hot → cold until same T & same avg KE.

Dissolving: ΔH_{soln} depends on relative strength of broken vs formed IMFs (breaking is endo > 0, forming is exo < 0).

Phase changes (T const): melt/vap = endo (+); freeze/cond = exo (–). H_2O: $\Delta H_{fus} = 6.0$ kJ/mol; $\Delta H_{vap} = 40.7$ kJ/mol.

Heating curve: sloped (heat warms, $q = mc\Delta T$) → flat (phase change, T const, $q = n\Delta H$). Vap plateau > fus plateau.

Enthalpy & Hess's Law

Hess's Law: sum ΔH across path. Reverse → negate. $\times n \to \times n$. Cancel identical intermediates on oppo. sides. Ex: A→B ($\Delta H_1 = 100$) + B→C ($\Delta H_2 = 50$) → A→C ($\Delta H = 150$ kJ).

Enthalpy: const P: $\Delta H = q_p$.Std state (°): 298 K, 1 atm, 1 M. $\Delta H^\circ_{rxn} = \Sigma n\Delta H^\circ_f(\text{products}) - \Sigma n\Delta H^\circ_f(\text{reactants})$. Pure elements in standard state: $\Delta H^\circ_f = 0$.

Bond Energy

BE = $\Delta H_{rxn} = \Sigma BE_{broken} - \Sigma BE_{formed}$. E to break (endo, +). **Exo:** strong bonds formed ($\Delta H < 0$). **Endo:** weak bonds formed ($\Delta H > 0$). **Exo:** $H_2 + F_2 \to 2HF$. Broken: 436+158. Formed: $2(567)$. $\Delta H = 594 - 1134 = -540$ kJ.

✓ FRQ: ΔH°_f if given; Hess for multi-step; signs (exo –, endo +); state heat-flow direction. Units always. Bond energies: Draw Lewis structures to count bonds.

Equilibrium (Unit 7, 7–9%)

K Expression & ICE Table

$K_c = [C]^c[D]^d/[A]^a[B]^b$ for $aA + bB \rightleftharpoons cC + dD$. $K > 1$: prod favored. $K < 1$: react favored. $K_{rev} = 1/K_{fwd}$. K_p uses partial pressures. **Pure s/l excluded.**

ICE: I (initial), C ($\pm x$ per stoich), E = I+C. Sub E into K, solve x (quadratic, or assume x negligible if $K \gg 1$ or $K \ll 1$).

5% rule: approx valid if $x/[A]_0 < 5\%$. (justifies dropping $\pm x$)

Coupled rxns: if rxn = step1 + step2, then $K_{net} = K_1 \cdot K_2$.

Rxn ×n: $K_{new} = K^n$. **Large K:** rxn goes to ~completion.

Q vs K & Le Chatelier

Q: same form as K, current concentration/pressures $Q < K$: shift → (right). $Q > K$: shift ← (left). $Q = K$: at equilibrium.

Le Chatelier — shift away from stress: +reactant → right; +product → left. ↑ T shifts *endo* direction (K changes). ↑ P (compress) → fewer-mol-gas side; volume opposite. Catalyst: faster (lowers E_a for both fwd & rev), no shift, no K change. **Inert gas** (const V): no shift (partial P unchanged).

Dilute w/ H_2O: shifts to side w/ more aq moles.

Dynamic eq: Forward & reverse rxn rates equal → in no net observable change (concentrations remain constant, NOT equal).

Solubility & K_{sp}

K_{sp}: $AgCl(s) \rightleftharpoons Ag^+ + Cl^- \to K_{sp} = [Ag^+][Cl^-]$.

Molar solubility s: e.g. $K_{sp} = s^2$ for AgCl, $4s^3$ for CaF_2.

Q_{ion} **vs** K_{sp}: $> \to$ precipitate; $< \to$ dissolves; $= \to$ saturated.

Common ion: add $Cl^- \to \uparrow Q_{ion} \to$ shift ← → ↓ solubility.

pH effect: insol bases more soluble in acid.

Comparing solubility: Only rank directly via K_{sp} if salts yield the same number of ions; otherwise, calculate s.

✓ FRQ: write K (omit s/l); ICE w/ x; solve + check 5%; Q vs K for direction; Le Chat for shifts; K_{sp} via Q_{ion}.

Acids & Bases (Unit 8, 11–15%)

✓ Strong A: $[H^+] = M$. $K_a \cdot K_b = K_w = 10^{-14}$. Equiv pH: 7 (S+S); > 7 (WA+SB); < 7 (SA+WB).

Strong vs Weak, K_a/K_b

Brønsted-Lowry: A donates H^+, B accepts; conj pair differs by 1 H^+. Amphoteric species (e.g., H_2O, HCO_3^-) act as A or B.

Lewis: A = e⁻ pair acceptor, B = donor.

Strong A (100% ioniz): HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$. Weak: all others (e.g., HF, carboxylic acids; K_a + ICE).

Strong B: Group 1 & 2 OH (e.g., $Ba(OH)_2 \to 2OH^-$).

Weak B: NH_3, amines, carboxylates (K_b + ICE).

$K_w = [H^+][OH^-] = 10^{-14}$. $pH = -\log[H^+]$. $pH + pOH = 14$. $pK_a + pK_b = 14$.

Stronger A = ↑ K_a, ↓ pK_a. → weaker conj base (↓ K_b, ↑ pK_b)

Salt: S A + W B → acidic; W A + S B → basic; S+S → neutral.

Polyprotic: $K_{a1} \gg K_{a2}$; usually only K_{a1} matters.

pH Calculations & Buffers

Strong A/B: $[H^+] = M$. **Weak A (HA):** $HA \rightleftharpoons H^+ + A^-$. $K_a = [H^+][A^-]/[HA]$. If $K_a \ll 0.05C$: $[H^+] = \sqrt{K_aC}$.

Weak B: $B + H_2O \rightleftharpoons BH^+ + OH^-$. $K_b = [BH^+][OH^-]/[B]$. If $K_b \ll 0.05C$: $[OH^-] = \sqrt{K_bC}$. $pH = 14 - pOH$.

Buffer: weak A + conj base (or weak B + conj acid).

H-H: $pH = pK_a + \log([A^-]/[HA])$. Effective $pH \approx pK_a \pm 1$. Max capacity at $pH = pK_a$ ($[A^-] = [HA]$).

Titration Curves

Equiv pt: mol H^+ = mol OH^-. S+S: $pH \approx 7$. WA+SB: > 7 (A^- basic). WB+SA: < 7 (BH^+ acidic). **Half-equiv:** $pH = pK_a$ ($[A^-] = [HA]$). **Indicator:** color near its pK_a.

⚠ Weak A approx valid if $x/C_0 < 5\%$. Curve: buffer → steep (equiv) → flat. Polyprotic: 1 drop per H^+.

Applications of Thermodynamics (Unit 9, 7–9%)

Gibbs Free Energy & Spontaneity

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ (T in K). If $\Delta G^\circ < 0 \to$ thermodynamic. favored; if slow/no rxn → it's under kinetic control (high E_a).

ΔH°	ΔS°	$\Delta G^\circ < 0$ at:	Driving Force
< 0 (–)	> 0 (+)	All T	Both
> 0 (+)	< 0 (–)	No T	Neither
> 0 (+)	> 0 (+)	High T	Entropy
< 0 (–)	< 0 (–)	Low T	Enthalpy

K link: $\Delta G^\circ = -RT\ln K$ (R = 8.314 J/mol·K). $K > 1 \to$ favored; $K < 1 \to$ not. If $\Delta G^\circ \approx 0$, then $K \approx 1$.

Electrochemistry: Redox & Cell Potentials

Galvanic (spont): anode = ox (–), cathode = red (+). $E^\circ_{cell} = E^\circ_{cath} - E^\circ_{anode}$ (use reduction potentials).

Notation: anode l salt bridge ll cathode. e^- flow: anode → cathode. $E^\circ > 0 \to$ spont; $< 0 \to$ electrolytic.

Nernst (25°C): $E = E^\circ - (0.0592/n)\log Q$. Q = conc/pressure. At eq: $E = 0$, $Q = K$.

Link: $\Delta G^\circ = -nFE^\circ$ (F = 96485 C/mol, n = mol e^- transferred). Combined: $\Delta G^\circ = -RT\ln K = -nFE^\circ$.

Electrolysis (non-spont): applied V forces rxn. Anode: ox (+ term); Cathode: red (– term). **Faraday:** mol e⁻ = (A × s) / 96485.

⚠ E° from tables = reductions. To use as ox: reverse + negate. Coeff's don't alter E° (intensive). Use Nernst when conc ≠ 1 M

Coupled rxns: Unfavorable rxns can be driven by coupling to favorable ones (must share an intermediate, overall $\Delta G^\circ < 0$).

✓ FRQ: half-rxns → E° from table → $E^\circ_{cell} = E^\circ_{cath} - E^\circ_{anode}$ → spont if > 0 → Nernst for non-std → $\Delta G^\circ = -nFE^\circ$.